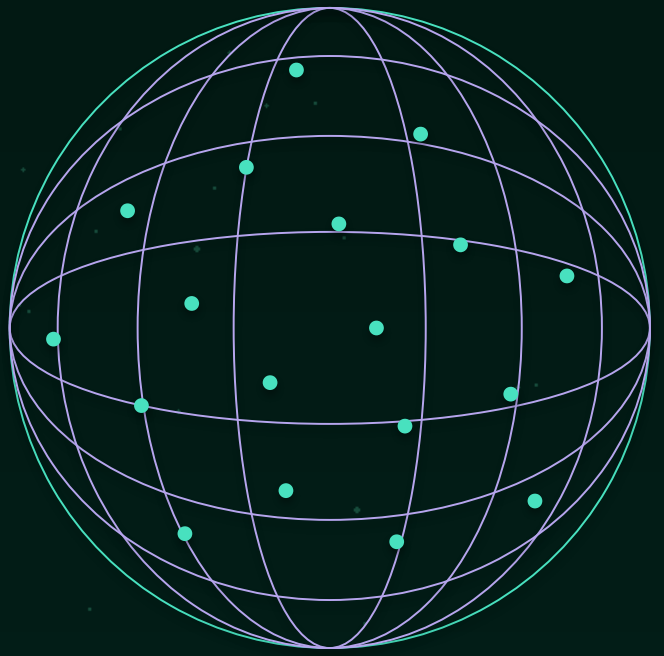
图溯

面向数字图像可信传播的来源认证与溯源方案

生成式水印 × 匿名登记 × 传播后核验



球形水印 · ICLR 2026

从内容生成，到传播后的可信核验

问题定义

生成图像的可追踪身份，应独立于附带文本存在



生成身份

谁的模型／哪条消息

传播变化

内容保存，附带说明丢失

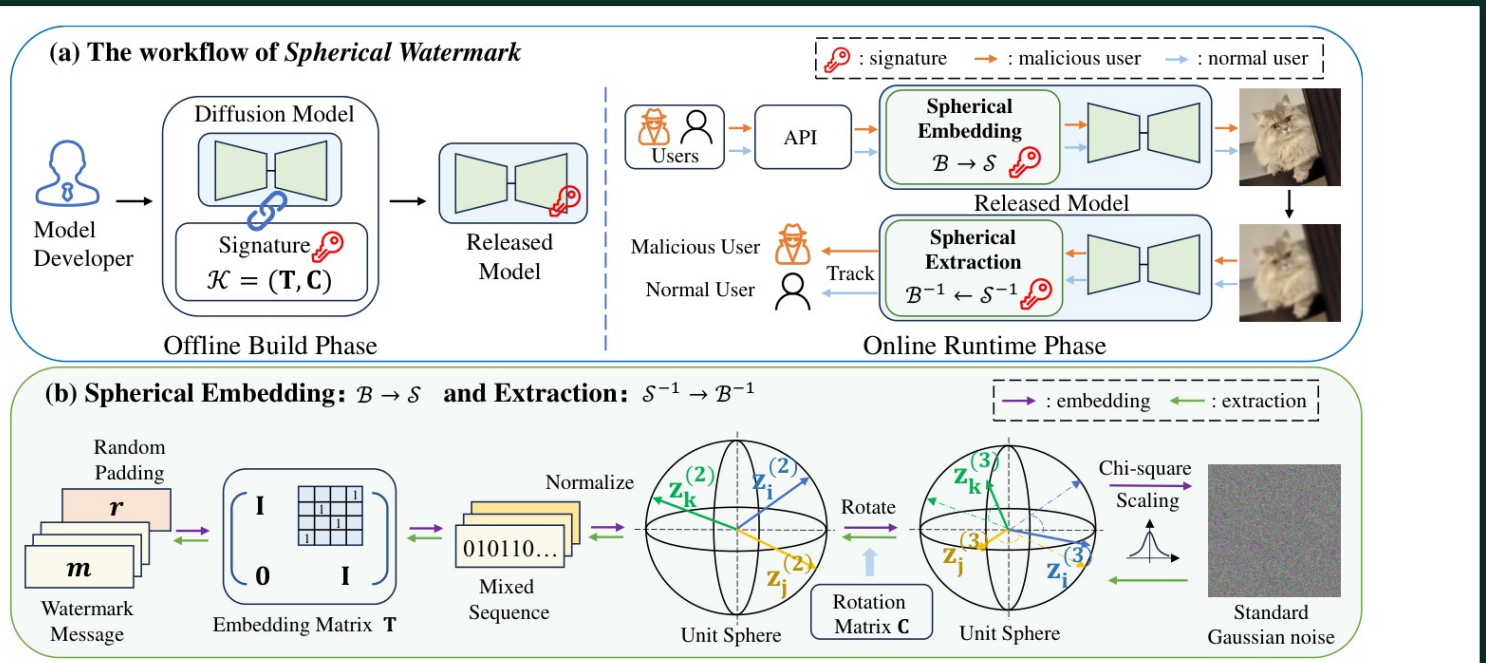
认证目标

恢复消息并验证关联

生成能力与传播能力提升，需要可核验的内容身份

研究定位

从后处理水印，到生成式潜在噪声水印



后处理路线

先获得图像，再对像素编码

生成式路线

在模型生成链路中嵌入信号

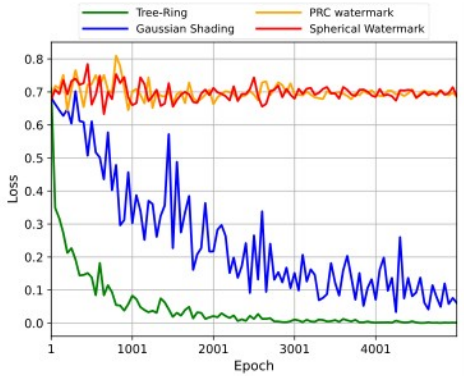
球形水印

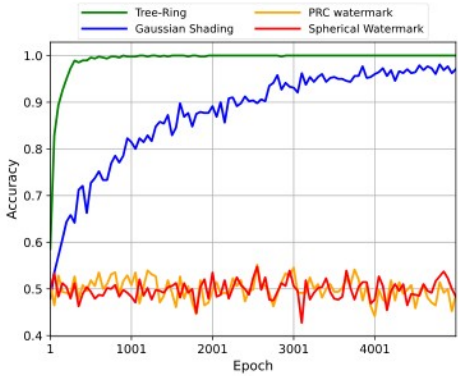
通过潜在噪声映射承载消息

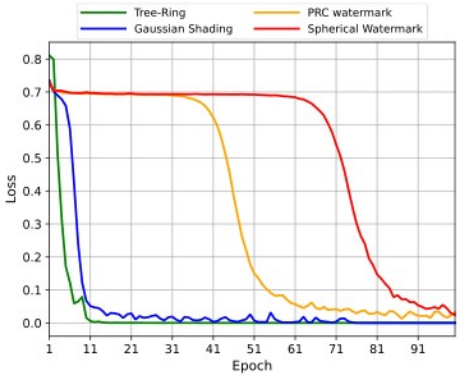
以潜在噪声承载身份信息，为生成过程提供主动标记

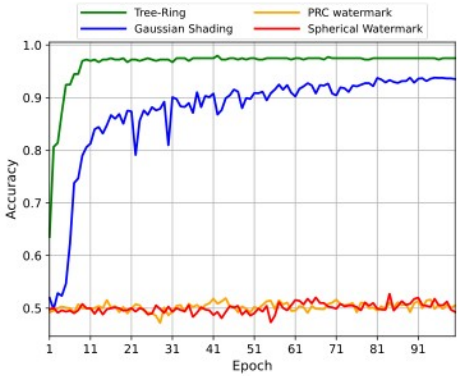
研究目标

分布扰动、视觉质量与消息恢复的联合评估









分布扰动

区分器表现与先验适配

感知质量

画质指标与视觉例图

恢复能力

比特准确率与检测误检

从分布、观感与恢复三个维度评价水印方法

关键痛点

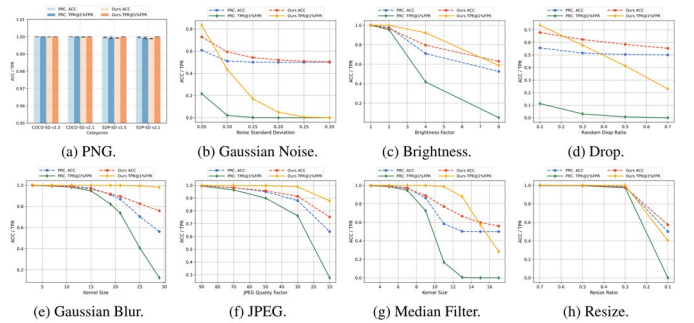
信息会分离，图像会变换，水印也有失败边界

画质扰动



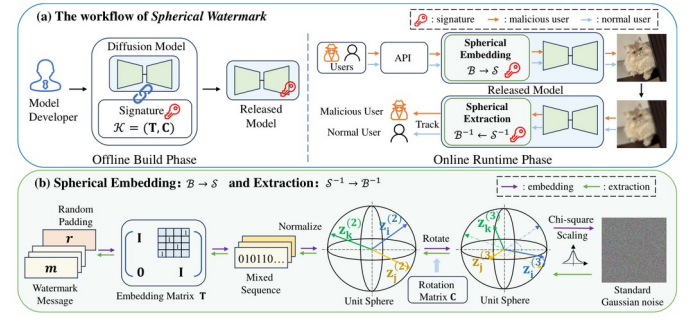
不应为身份标识明显牺牲图像观感

传播变换



压缩、缩放与裁剪可能损伤恢复能力

身份与权利



检出标识后仍需核对登记与授权材料

需要“算法保护＋登记关联＋人工复核”的完整闭环

总体技术路线

生成端嵌入身份，接收端直接上传核验

01 生成时建立身份

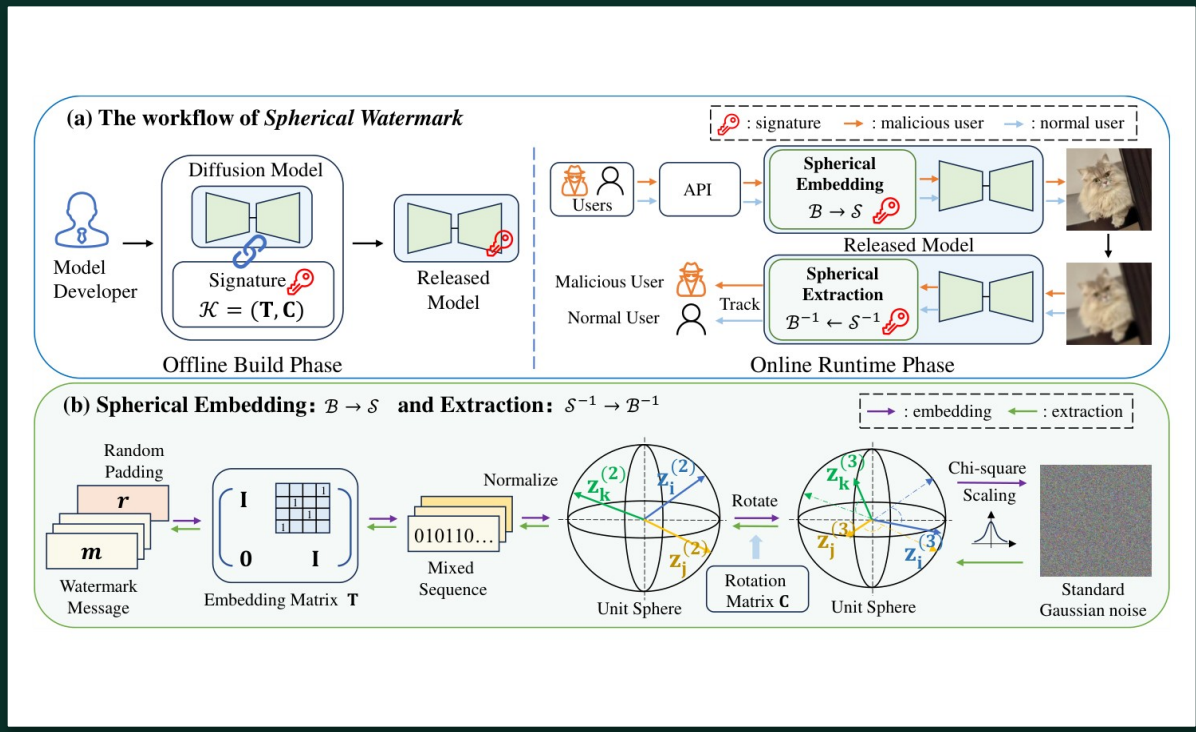
作品编号映射为噪声，再输入扩散模型。

02 传播后恢复标识

上传收到的图片，反演并提取匿名编号。

03 关联声明与人工复核

显示登记线索；不确定结果进入人工审核。



无需用户预先说明图片经历过哪些传播操作

可逆编码

重复消息与二进制矩阵混合

$$\mathbf{T} = \begin{bmatrix} \mathbf{I}_{l_{Nm}} & \mathbf{R} \\ \mathbf{0} & \mathbf{I}_{l_r} \end{bmatrix}, l_{Nm} = N \times l_m. \tag{6}$$

Algorithm 1 Construction of Binary Matrix $\mathbf{R}$

Require: Positive integers N, l_m, l_r, s such that $l_r \geq N \times s$
Ensure: Binary matrix $\mathbf{R} \in \{0,1\}^{l_{Nm} \times l_r}$, Indices Set P

Initialize $\mathbf{R} \leftarrow \mathbf{0}^{N \times l_m \times l_r}$
$P \leftarrow \emptyset$
for $j = 1$ to l_m **do**
  $\pi \leftarrow \text{RandomPermutation}([1, 2, \ldots, l_r])$
  $TMP \leftarrow \pi[1 : N \times s]$
  for $i = 1$ to N **do**
    $G \leftarrow TMP[(i-1) \times s + 1 : i \times s]$
    $\mathbf{R}[i, j, G] \leftarrow 1$
    $P \leftarrow P \cup \{(i, j, G)\}$
  end for
end for
Return $\text{Reshape}(\mathbf{R}, (l_{Nm}, l_r))$, P

01 单位块与稀疏随机块

保持 T 可逆，同时混合随机填充

02 重复比特分组

通过独立填充子集支持恢复分析

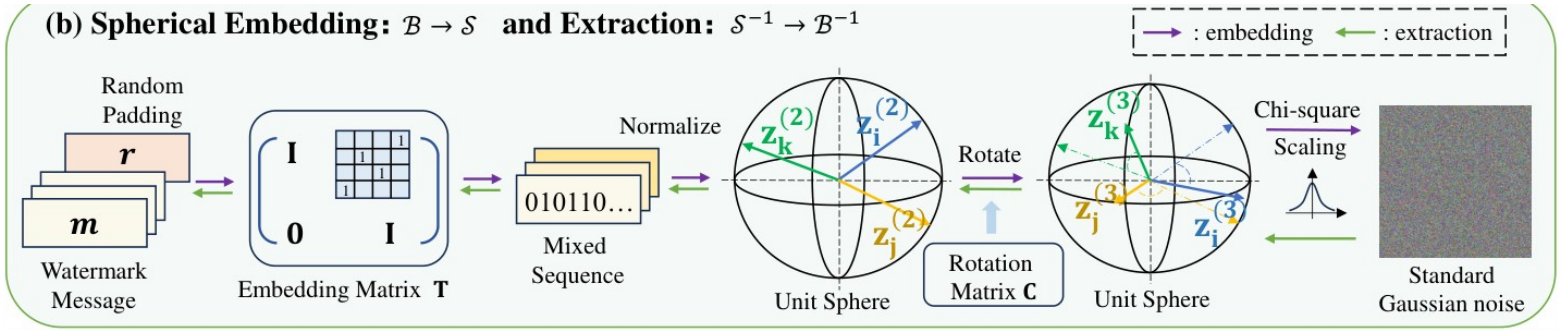
x = [m, ..., m, r]

z(1) = T x (GF(2))

T 可逆，重复消息与随机填充共同支持编码和恢复

核心方法二

球面映射与随机半径：构造生成端潜在噪声



$$\mathbf{v} = 2\,\mathbf{z}^{(1)} - \mathbf{1}, \mathbf{z}^{(2)} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}, \mathbf{z}^{(3)} = \mathbf{C}\,\mathbf{z}^{(2)},$$
$$\text{draw } r \text{ such that } r^2 \sim \chi^2(l_x), \mathbf{z}_w = r\,\mathbf{z}^{(3)}. \tag{10}$$

01 单位球面

二进制信号归一化

02 正交旋转

由签名矩阵 C 变换

03 随机半径

平方半径服从卡方分布

采用球形水印的生成适配方法，而非在成图上叠加可见纹理

提取机制

噪声反演、逆变换与消息恢复

$$\hat{\mathbf{z}}^{(2)} = \mathbf{C}^{-1}\,\hat{\mathbf{z}}_T, \hat{\mathbf{z}}^{(1)} = \mathrm{round}\left(\frac{\hat{\mathbf{z}}^{(2)}+\mathbf{1}}{2}\right), \hat{\mathbf{x}} = \mathbf{T}^{-1}\,\hat{\mathbf{z}}^{(1)}, \tag{13}$$

传播图像

VAE 编码

噪声反演

近似 ODE 求解

逆映射

旋转逆变换与解混合

消息恢复

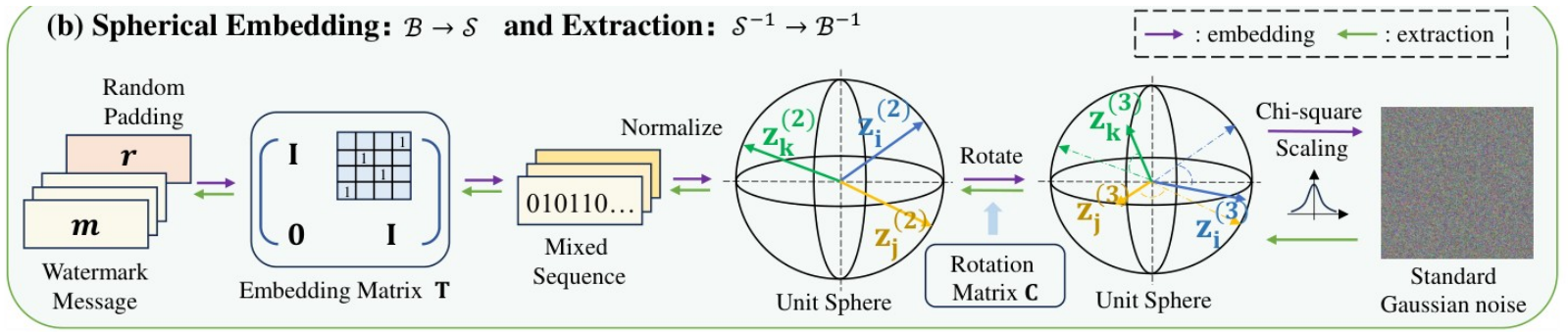
重复比特多数投票

误差来源：传播变换 → 潜在编码误差 → 反演误差 → 比特翻转

恢复链路的每一层误差，都可能传导到最终消息

签名构建

离线签名 K=(T,C)，支撑在线嵌入与提取



K = (T, C)

签名固定复用
无逐图加密流程
仍须保密管理

离线签名

T： 可逆二进制矩阵

正交构造

C： 正交旋转矩阵

在线运行

固定签名嵌入与提取

离线固定签名连接生成端与核验端

复现状态

用可复查的软件证据说明完成度

当前原型

图片上传与登记

后处理水印接口

压缩／缩放／裁剪

来源查询与历史记录

软件验证

13 / 13

现有工作流单元测试通过
包括持久化与错误语义

错误处理与持久化



SPHERICAL WATERMARK: ENCRYPTION-FREE, LOSSLESS WATERMARKING FOR DIFFUSION MODELS

Xiaoxiao Hu[1], **Jiaqi Jin**[1], **Sheng Li**[1], **Wanli Peng**[2], **Xinpeng Zhang**[1], **Zhenxing Qian**[1,*]
[1]Fudan University, [2]China Agricultural University
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{lisheng,zhangxinpeng,zxqian}@fudan.edu.cn
wlpeng@cau.edu.cn

ABSTRACT

Diffusion models have revolutionized image synthesis but raise concerns around content provenance and authenticity. Digital watermarking offers a means of tracing generated media, yet traditional schemes often introduce distributional shifts and degrade visual quality. Recent lossless methods embed watermark bits directly into the latent Gaussian prior without modifying model weights, but still require per-image key storage or heavy cryptographic overhead. In this paper, we introduce **Spherical Watermark**, an encryption-free and lossless watermarking framework that integrates seamlessly with diffusion architectures. First, our binary embedding module mixes repeated watermark bits with random padding to form a high-entropy code. Second, the spherical mapping module projects this code onto the unit sphere, applies an orthogonal rotation, and scales by a chi-square-distributed radius to recover exact multivariate Gaussian noise. We theoretically prove that the watermarked noise distribution preserves the target prior up to third-order moments, and empirically demonstrate that it is statistically indistinguishable from a standard multivariate normal distribution. Adopting Stable Diffusion, extensive experiments confirm that Spherical Watermark consistently preserves high visual fidelity while simultaneously improving traceability, computational efficiency, and robustness under attacks, thereby outperforming both lossy and lossless approaches.

1 INTRODUCTION

Diffusion models have demonstrated transformative potential in creative applications (Rombach et al., 2022; Sahoo et al., 2024), but also raise concerns over authenticity and ownership (Craver et al., 1997; Grinbaum & Adomaitis, 2022). Malicious users can exploit them to fabricate images and spread disinformation, eroding public trust and creating legal and ethical challenges. As governments and platforms face mounting pressure to address harmful content (Biden, 2023; Wiggers, 2023), reliable provenance mechanisms are urgently needed to trace and identify malicious actors.

Image watermarking offers a promising direction by embedding imperceptible identifiers into images However, traditional schemes alter the data distribution and degrade visual fidelity, whether operating in the spatial (Li et al., 2009; Bender et al., 1995) or frequency (Al-Haj, 2007; Navas et al., 2008) domain. Additionally, some approaches inject watermarks by training or fine-tuning generative models For example, Fernandez et al. (Fernandez et al., 2023) fine-tune the Stable Diffusion (Rombach et al.,

从图片保护到传播后核验，以统一接口连接模型与应用

质量验证

控制生成配置，比较感知质量与局部细节



控制变量

模型、步数、分辨率与种子

配对比较

相同提示词与一致采样配置

视觉核查

纹理、文字、主体与局部细节

同一生成配置与统一数据，是公平比较的基础

实施条件

论文、代码、算力与登记安全共同支撑落地



SPHERICAL WATERMARK: ENCRYPTION-FREE, LOSSLESS WATERMARKING FOR DIFFUSION MODELS

Xiaoxiao Hu[1], **Jiaqi Jin**[1], **Sheng Li**[1], **Wanli Peng**[2], **Xinpeng Zhang**[1], **Zhenxing Qian**[1,*]
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wlpeng@cau.edu.cn

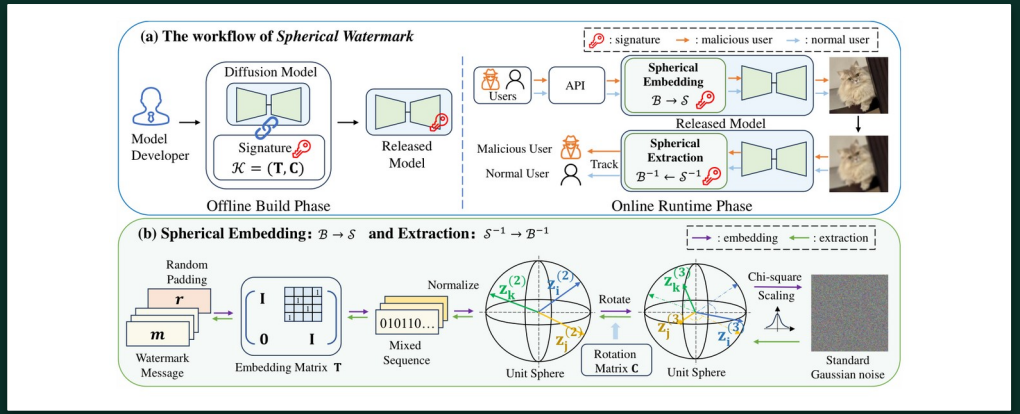
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研究基础

Spherical Watermark · ICLR 2026

对比路线

Tree-Ring / Gaussian Shading / PRC

计算资源

GPU 生成与反演
记录模型与采样配置

数据资源

自有或授权素材
无水印负样本

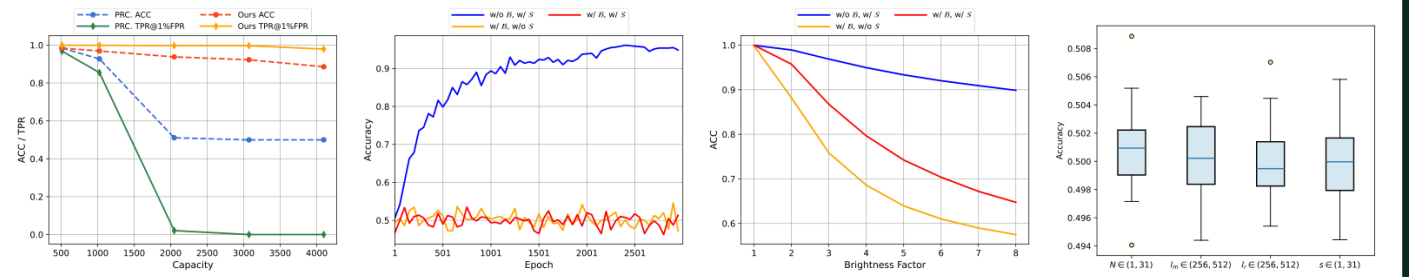
安全边界

受控签名与登记访问
不嵌入个人敏感信息

技术壁垒体现在稳定复现、低误检核验和可审计流程的结合

参数消融

重复次数、容量与反演配置的权衡



(a) Ablation on l_m. (b) Ablation on Modules. (c) Ablation on Modules. (d) Ablation on Settings.

消息长度

容量与恢复难度

重复次数

冗余开销与鲁棒性

反演设置

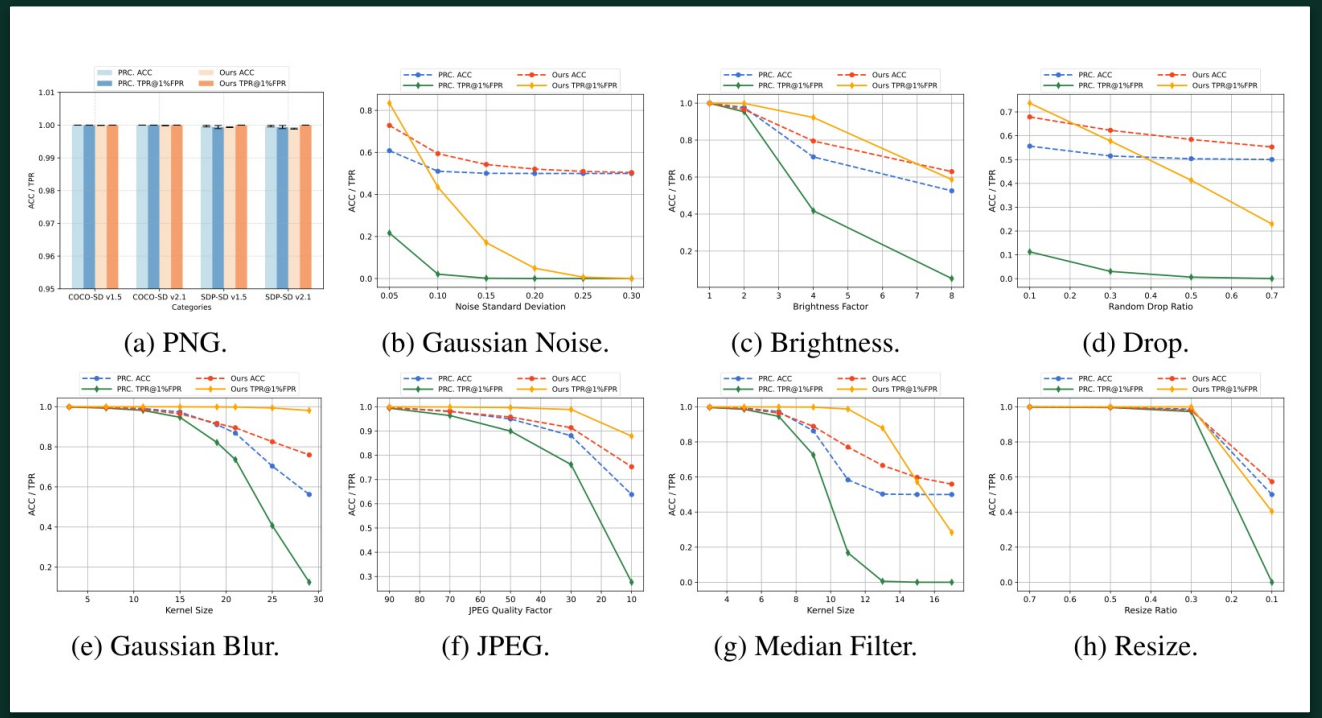
采样成本与误差积累

消融实验逐项改变变量，避免把参数效应误判为方法收益。

在容量、冗余、计算成本与恢复性能之间寻找适配点

测试与验证

论文证据说明潜力，项目实测检验可用性



(a) PNG. (b) Gaussian Noise. (c) Brightness. (d) Drop.

(e) Gaussian Blur. (f) JPEG. (g) Median Filter. (h) Resize.

01

画质与观感

质量指标＋局部细节

02

传播恢复

压缩／缩放／裁剪

03

误检与混淆

负样本＋错误签名

04

应用体验

正确判断＋任务耗时

同时报告成功、失败与无法确认；不只展示成功样例

方案比较

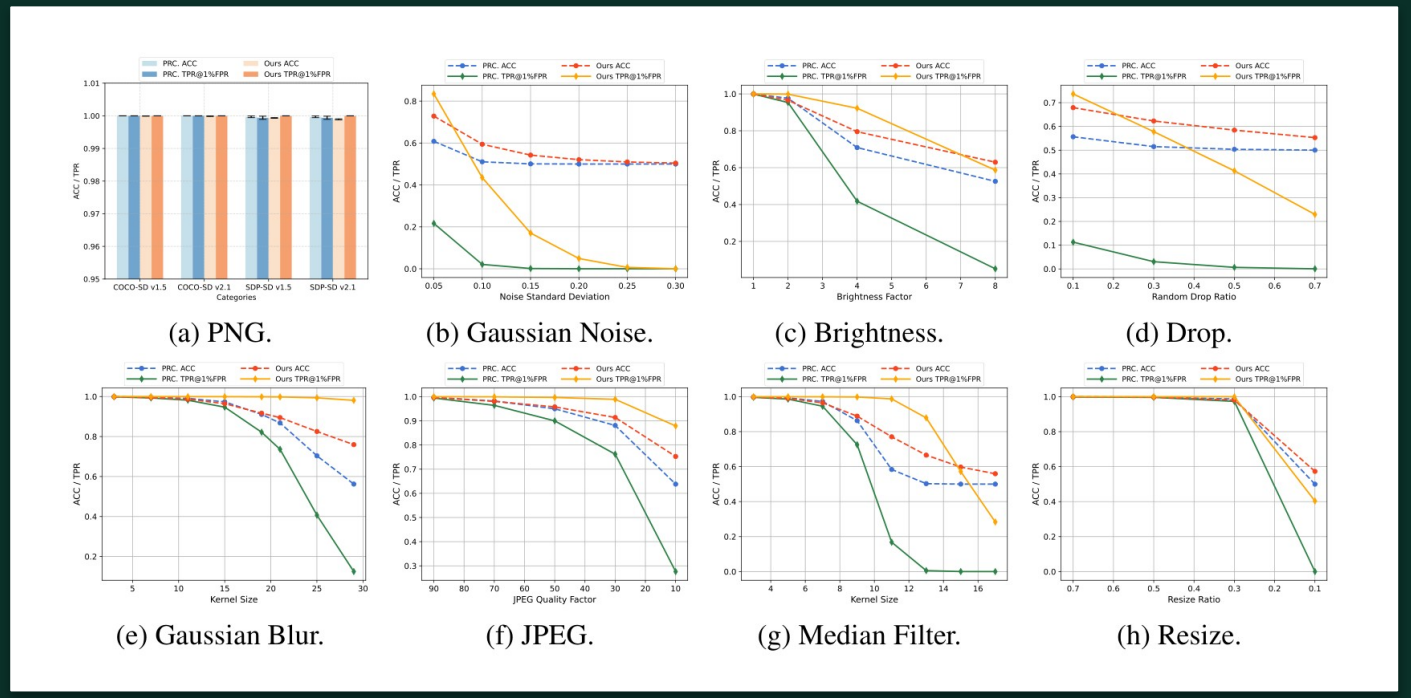
区分技术能力与应用能力，按同一任务对比

可见署名	文件元数据	后处理水印	球形水印	图溯方案
成图叠字	附带信息	修改已成图	生成噪声嵌入	**身份＋登记关联**
可直接看见	观感不变	适用上传图	生成链路适配	**收到图即可核验**
可能被覆盖	可能被移除	性能依模型	依赖图像反演	**失败进入复核**
需另存声明	需兼容读取	仍需登记关联	论文技术基础	**登记与核验闭环**

在同一来源查找任务下，比较保护能力与使用成本

可靠性评估

正样本、负样本与开放失败共同构成验证边界



(a) PNG. (b) Gaussian Noise. (c) Brightness. (d) Drop.

(e) Gaussian Blur. (f) JPEG. (g) Median Filter. (h) Resize.

正样本

水印图／传播变换图

负样本

无水印图／错误签名

开放失败

重绘、强裁剪与域外图像

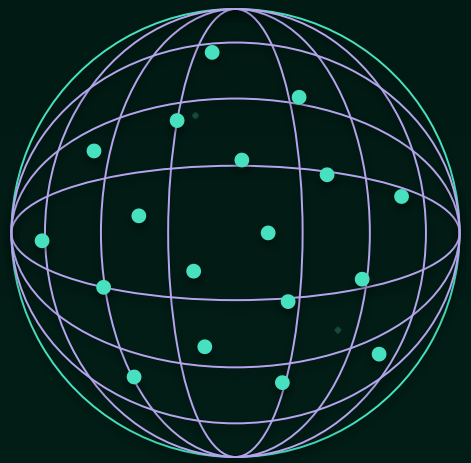
既评估可恢复性，也评估误检、混淆与无法确认

社会价值

为图像使用者提供可核实的来源线索

来源认证

辅助核对图片是否匹配
预先登记的生成身份



可信溯源

把恢复的标识关联到
受控登记与历史记录

权益保护

为署名、授权核对与争议
提供可复查的辅助线索

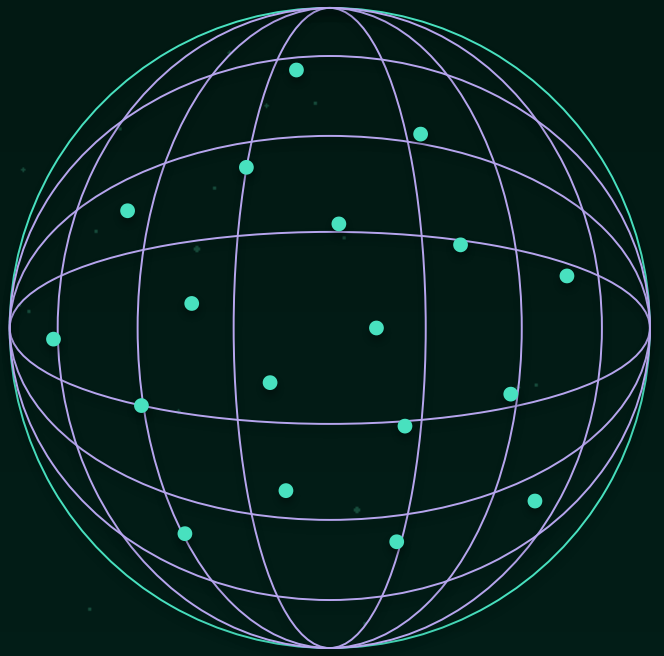
责任边界

不把检出标识等同于
事实真实或法律确权

让身份线索随图传播，让使用决策有据可查

让每一次传播，都保留可核验的身份线索

来源可认证
传播可追溯
权益有线索



球形水印 · ICLR 2026

从内容生成，到传播后的可信核验